

*Computer Science BSc; Analysis-1*  
*Theorems to be proved in the Advanced Level Part of the Theoretical Exam*  
*(40 theorems)*  
*2025/2026 Spring*

The abbreviation "AN-1" denotes the following reference:

*Analysis 1 lecture schemes, written by István Csörgő*

The abbreviation "AN-2" denotes the following reference:

*Analysis 2 lecture schemes, written by István Csörgő*

They can be found in the Digital Library of the Faculty of Informatics or they can be opened from CANVAS

1. Triangle Inequalities in  $\mathbb{R}$   
(*"AN-1 " p. 17*)
2. Theorem about the connection between the convergency and the ordering relations  
(*"AN-1 " p. 32*)
3. Theorem about the connection between the convergent and the bounded sequences  
(*"AN-1 " p. 34*)
4. Five theorems (Th1-Th5) in connection with zero sequences  
(*"AN-1 " pp. 35-36*)
5. Operations with convergent sequences: addition  
(*"AN-1 " p. 37*)
6. Operations with convergent sequences: multiplication  
(*"AN-1 " p. 38*)
7. Operations with convergent sequences: reciprocal  
(*"AN-1 " pp. 38-39*)
8. The Sandwich Theorem  
(*"AN-1 " p. 41*)
9. Theorem about the convergence of geometric sequences  
(*"AN-1 " p. 42*)
10. Theorem about the convergence of  $\sqrt[n]{a}$  and of  $\sqrt[n]{n}$   
(*"AN-1 " pp. 42-43*)
11. Theorem about the convergence of  $n^k \cdot q^n$  and of  $\frac{n^k}{a^n}$   
(*"AN-1 " pp. 43-44*)

12. Theorem about the convergence of  $\frac{x^n}{n!}$   
(*"AN-1 " p. 44*)
13. Theorem about the convergence of a monotonically increasing sequence  
(*"AN-1 " p. 49*)
14. The sequence  $\left( \left( 1 + \frac{1}{n} \right)^n \right)$  is convergent  
(*"AN-1 " p. 50*)
15. Theorem about the connection between the addition and the infinite limit of a sequence  
(*"AN-1 " p. 54*)
16. Theorem about the connection between the multiplication and the infinite limit of a sequence  
(*"AN-1 " p. 56*)
17. The theorem about the convergence and the sum of the geometric series  
(*"AN-1 " p. 62*)
18. The Zero-sequence Test (two directions)  
(*"AN-1 " p. 63*)
19. The theorem about the Cauchy's Convergence Test for series  
(*"AN-1 " p. 64*)
20. The theorem about the Comparison Tests (Major Test, Minor Test)  
(*"AN-1 " p. 65*)
21. The theorem about the convergence of Leibniz-type series Leibniz Criterion)  
(*"AN-1 " p. 68*)
22. The Root Test (+ some examples for the indeterminate case)  
(*"AN-1 " pp. 74-75*)
23. Theorem about the convergence set of a power series (using Root Test)  
(*"AN-1 " p. 87*)
24. The function  $\exp$  is well-defined for any  $x \in \mathbb{K}$   
(*"AN-1 " pp. 94-95*)
25. The connection between  $\exp$ ,  $\cosh$ ,  $\sinh$  and its important consequences  
(*"AN-1 " pp. 96-97*)
26. The connection between  $\exp$ ,  $\cos$ ,  $\sin$  (Euler's Identity) and its important consequences  
(*"AN-1 " pp. 96-97*)
27. Addition Formula of the exponential function  
(*"AN-1 " p. 97*)

28. Addition Formula of the cosine function  
(*"AN-1 " p. 98*)
29. The estimation of error for the series of  $e = \exp(1)$   
(*"AN-1 " p. 101*)
30.  $e \notin \mathbb{Q}$   
(*"AN-1 " p. 102*)
31. The Theorem about Transference Principle for limit of functions  
(*"AN-1 " pp. 107-108*)
32. The basic limits  $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ ,  $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$ ,  $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$   
(*"AN-1 " pp. 124-125*)
33. Theorem about the characterization of closed sets with convergent sequences  
(*"AN-2 " p. 6, the proof was taught on the lectures IV. 30. and V. 1.*)
34. Theorem about the connection between the compact sets and the closed and bounded sets  
(*"AN-2 " p. 7*)
35. Theorem about the minimal and maximal elements of a compact set  
(*"AN-2 " pp. 7-8*)
36. Theorem about the compactness of the image  
(*"AN-2 " p. 9*)
37. The minimax theorem of Weierstrass  
(*"AN-2 " p. 9*)
38. Prove that the function  $f(x) = \frac{1}{x}$  ( $x > 0$ ) is continuous but it is not uniformly continuous  
(*"AN-2 " p. 10 and lecture*)
39. The Intermediate Value Theorem (Bolzano's Theorem)  
(*"AN-2 " p. 11*)
40. The real exponential function is strictly increasing  
(*"AN-2 " p. 13*)